EDEXCEL INTERNATIONAL GCSE MATH (4MA1) • UNIT 8.01

C Histograms, Cumulative Frequency & Conditional Probability

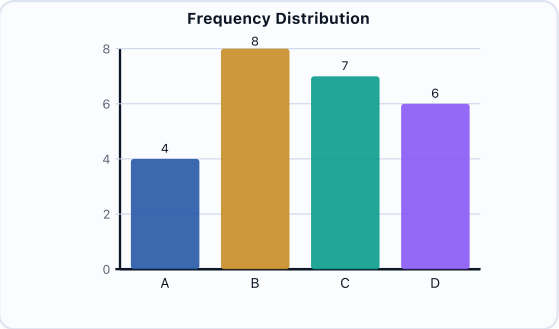
PRACTICE WORKSHEET

25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

Histograms, Cumulative Frequency & Conditional Probability Practice Exercises

1 MCQ

The coordinate diagram shows the parabolic curve $f_1(x) = x^2 - 20x + 118$ in the domain of Histograms, Cumulative Frequency & Conditional Probability. By completing the square $f_1(x) = (x - h)^2 + k$, find the sum $k + f_1(12)$.



A 50

B 22

C 38

D 40

2 MCQ

A mathematical model in Histograms, Cumulative Frequency & Conditional Probability is defined by the cubic function $y_2(x) = 3x^3 - 10x^2 + 15x - 15$. Calculate the exact gradient of the tangent to the curve at the point where $x = 3$.

A 36

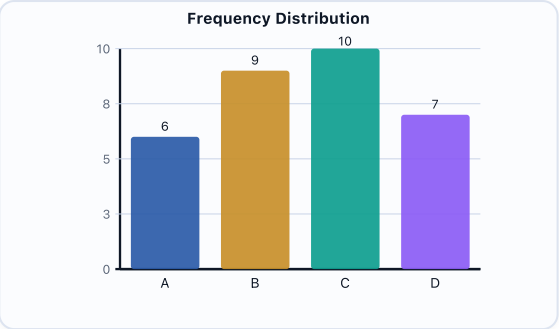
B 96

C 21

D 22

3 MCQ

In Histograms, Cumulative Frequency & Conditional Probability, two displacement vectors are given by $\mathbf{u}_3 = \langle 10, -8 \rangle$ and $\mathbf{v}_3 = \langle 9, 7 \rangle$. Compute the exact scalar value of $\mathbf{u}_3 \cdot \mathbf{v}_3 + \|\mathbf{u}_3\|^2$.



A 34

B 198

C 164

D 310

4 MCQ

The linear growth profile shown in the diagram for Histograms, Cumulative Frequency & Conditional Probability follows an arithmetic progression with first term $a_1 = 11$ and common difference $d = 6$. Find the exact sum S_{15} of the first 15 terms.

A 95

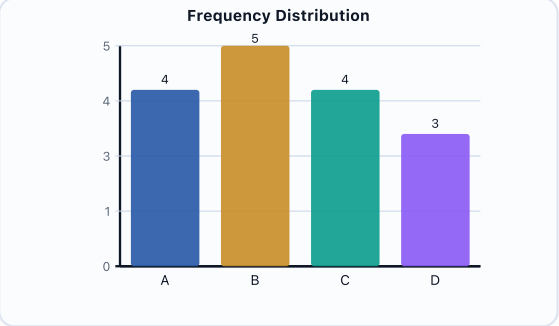
B 885

C 795

D 1425

5 MCQ

In Histograms, Cumulative Frequency & Conditional Probability, evaluate the exact definite integral $I_5 = \int_0^2 (18x^2 + 40x)\,\mathrm{d}x$.



A 152

B 168

C 65

D 128

6 MCQ

In Histograms, Cumulative Frequency & Conditional Probability, consider the quadratic function $f_6(x) = x^2 - 22x + 211$. Express $f_6(x)$ in the completed-square form $(x-h)^2 + k$ and compute the exact value of $k + f_6(18)$.

A 229

B 240

C 139

D 222

7 MCQ

The graph illustrates the cubic model $y_7(x) = 3x^3 - 11x^2 + 33x - 22$ studied in Histograms, Cumulative Frequency & Conditional Probability. Find the instantaneous gradient $\frac{dy}{dx}$ of the tangent to the curve at $x = 8$.

A 609

B 433

C 400

D 83

8 MCQ

In Histograms, Cumulative Frequency & Conditional Probability, two displacement vectors are given by $\mathbf{u}_8 = \langle 11, -7 \rangle$ and $\mathbf{v}_8 = \langle 8, 5 \rangle$. Compute the exact scalar value of $\mathbf{u}_8 \cdot \mathbf{v}_8 + \|\mathbf{u}_8\|^2$.

A 53

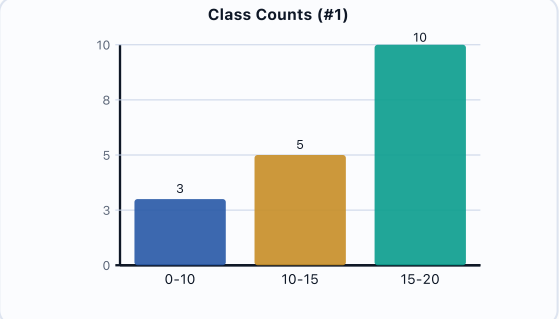
B 170

C 223

D 293

9 MCQ

In a grouped frequency table (#1), the class interval $10 < t \le 15$ contains 5 observations. Calculate the frequency density plotted on a histogram for this class.



A 25

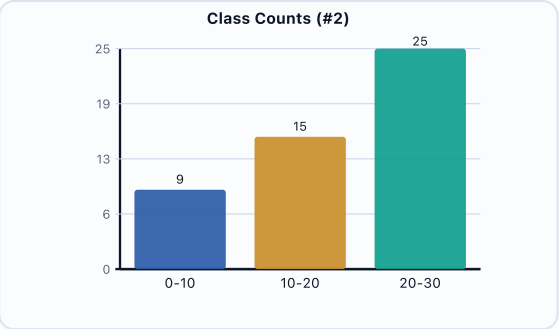
B 1.00

C 1

D 2.0

10 MCQ

In a grouped frequency table (#2), the class interval $10 < t \leq 20$ contains 15 observations. Calculate the frequency density plotted on a histogram for this class.



A 0.67

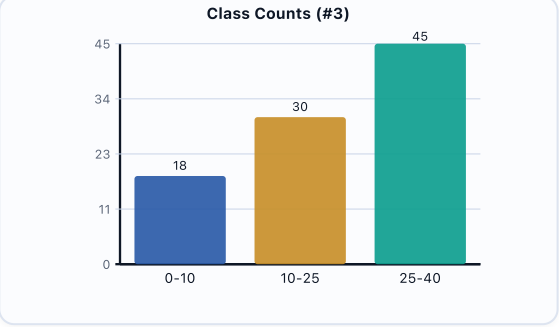
B 1.5

C 150

D 3.0

11 MCQ

In a grouped frequency table (#3), the class interval $10 < t \leq 25$ contains 30 observations. Calculate the frequency density plotted on a histogram for this class.



A 2

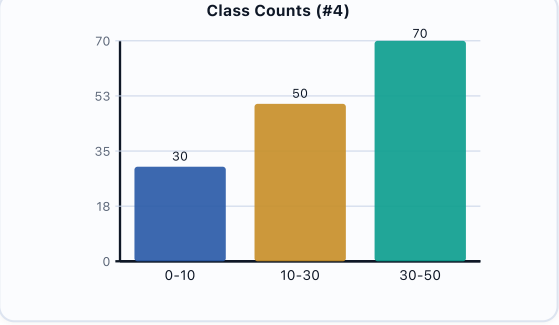
B 4.0

C 450

D 0.50

12 MCQ

In a grouped frequency table (#4), the class interval $10 < t \leq 30$ contains 50 observations. Calculate the frequency density plotted on a histogram for this class.



A 5.0

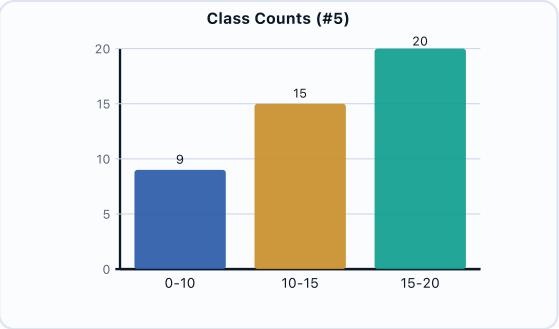
B 2.5

C 0.40

D 1000

13 MCQ

In a grouped frequency table (#5), the class interval $10 < t \leq 15$ contains 15 observations. Calculate the frequency density plotted on a histogram for this class.



A 3

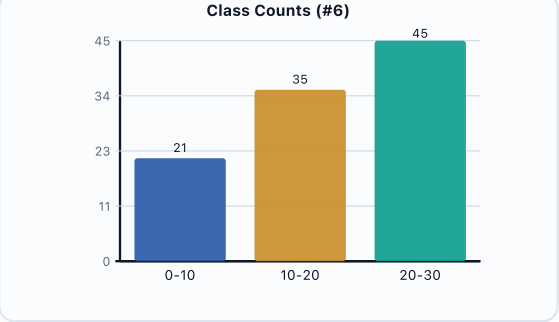
B 6.0

C 0.33

D 75

14 MCQ

In a grouped frequency table (#6), the class interval $10 < t \leq 20$ contains 35 observations. Calculate the frequency density plotted on a histogram for this class.



A 7.0

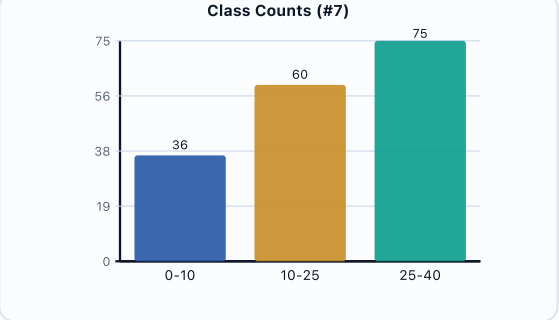
B 3.5

C 0.29

D 350

15 MCQ

In a grouped frequency table (#7), the class interval $10 < t \leq 25$ contains 60 observations. Calculate the frequency density plotted on a histogram for this class.



A 0.25

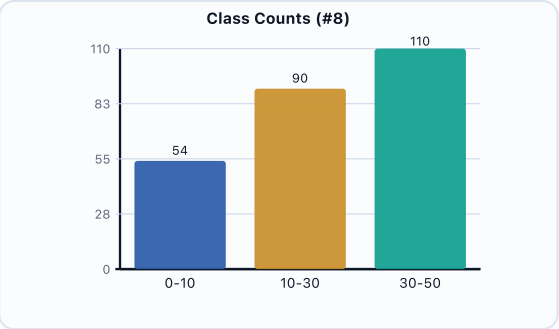
B 4

C 900

D 8.0

16 MCQ

In a grouped frequency table (#8), the class interval $10 < t \leq 30$ contains 90 observations. Calculate the frequency density plotted on a histogram for this class.



A 4.5

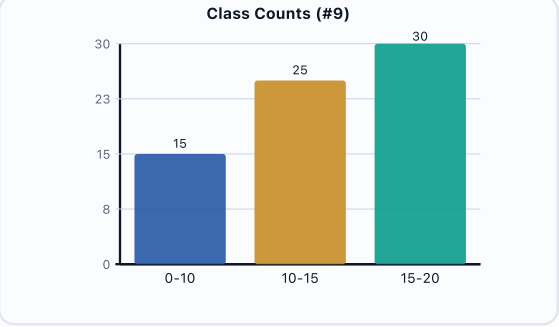
B 0.22

C 1800

D 9.0

17 MCQ

In a grouped frequency table (#9), the class interval $10 < t \leq 15$ contains 25 observations. Calculate the frequency density plotted on a histogram for this class.



A 0.20

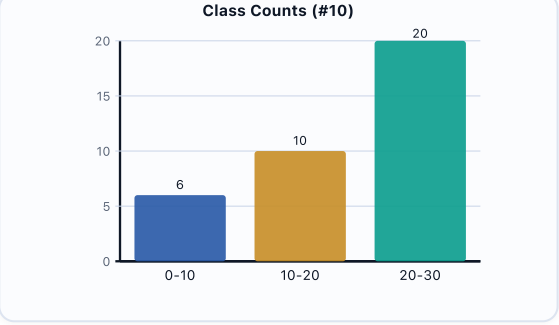
B 10.0

C 125

D 5

18 MCQ

In a grouped frequency table (#10), the class interval $10 < t \leq 20$ contains 10 observations. Calculate the frequency density plotted on a histogram for this class.



A 100

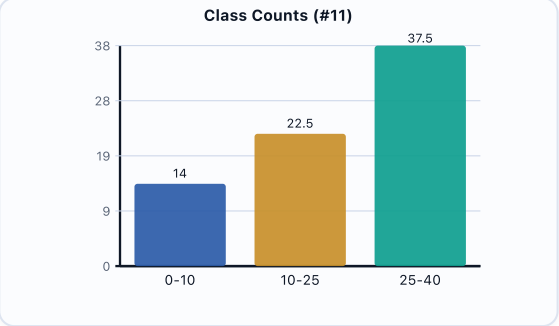
B 2.0

C 1.00

D 1

19 MCQ

In a grouped frequency table (#11), the class interval $10 < t \leq 25$ contains 22.5 observations. Calculate the frequency density plotted on a histogram for this class.



A 337.5

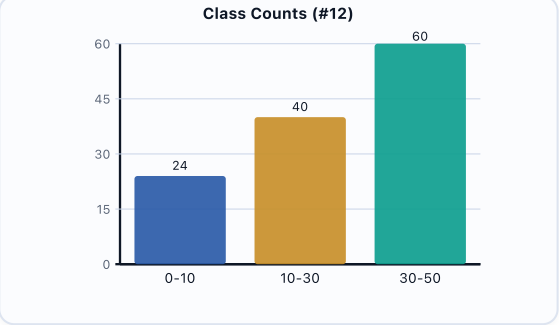
B 0.67

C 1.5

D 3.0

20 MCQ

In a grouped frequency table (#12), the class interval $10 < t \leq 30$ contains 40 observations. Calculate the frequency density plotted on a histogram for this class.



A 800

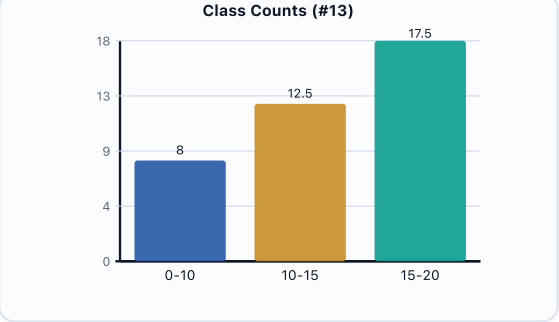
B 2

C 0.50

D 4.0

21 MCQ

In a grouped frequency table (#13), the class interval $10 < t \leq 15$ contains 12.5 observations. Calculate the frequency density plotted on a histogram for this class.



A 2.5

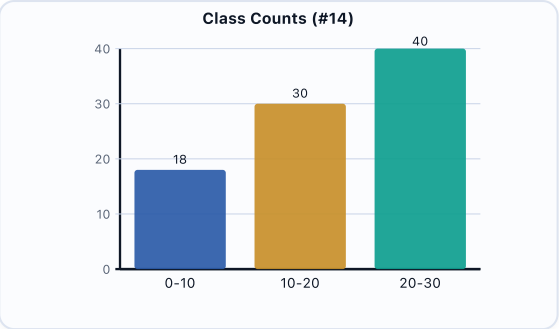
B 0.40

C 5.0

D 62.5

22 MCQ

In a grouped frequency table (#14), the class interval $10 < t \leq 20$ contains 30 observations. Calculate the frequency density plotted on a histogram for this class.



A 6.0

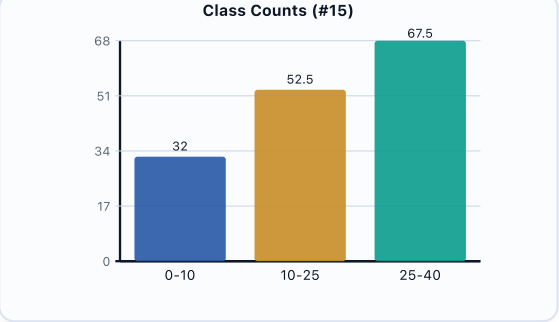
B 3

C 300

D 0.33

23 MCQ

In a grouped frequency table (#15), the class interval $10 < t \leq 25$ contains 52.5 observations. Calculate the frequency density plotted on a histogram for this class.



A 7.0

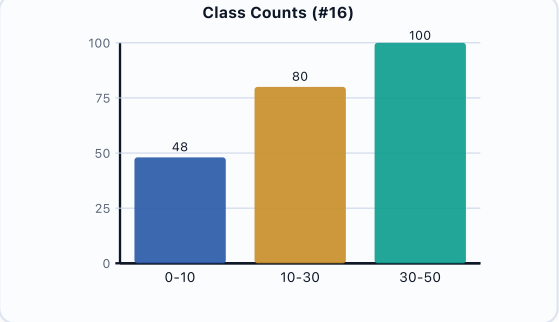
B 0.29

C 787.5

D 3.5

24 MCQ

In a grouped frequency table (#16), the class interval $10 < t \leq 30$ contains 80 observations. Calculate the frequency density plotted on a histogram for this class.



A 0.25

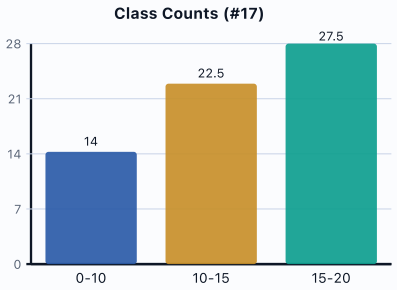
B 4

C 8.0

D 1600

25 MCQ

In a grouped frequency table (#17), the class interval $10 < t \leq 15$ contains 22.5 observations. Calculate the frequency density plotted on a histogram for this class.



A 112.5

B 9.0

C 4.5

D 0.22